# Ex: 2 - Data Size Converter (Java Packages)

Aim: To create a Java program using two packages (**converter** and **main**) that converts a given data size into Bytes, KB, MB, GB and TB.

## Program (1) Package: converter

*File: converter/DataSizeConverter.java*

```
package converter;

public class DataSizeConverter {
    private double bytes;

    public DataSizeConverter(double value, String unit) {
        switch (unit.toLowerCase()) {
            case "kb":
                this.bytes = value * 1024;
                break;
            case "mb":
                this.bytes = value * 1024 * 1024;
                break;
            case "gb":
                this.bytes = value * 1024 * 1024 * 1024;
                break;
            case "tb":
                this.bytes = value * 1024 * 1024 * 1024 * 1024;
                break;
            default: // bytes
                this.bytes = value;
        }
    }

    public double toKB() { return bytes / 1024; }
    public double toMB() { return bytes / (1024 * 1024); }
    public double toGB() { return bytes / (1024 * 1024 * 1024); }
    public double toTB() { return bytes / (1024L * 1024 * 1024 * 1024); }
    public double toBytes() { return bytes; }
}
```

## Program (2) Package: main

*File: main/ConverterApp.java*

```
package main;

import converter.DataSizeConverter;
import java.util.Scanner;

public class ConverterApp {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter value: ");
        double value = sc.nextDouble();

        System.out.print("Enter unit (Bytes/KB/MB/GB/TB): ");
        String unit = sc.next();

        DataSizeConverter converter = new DataSizeConverter(value, unit);

        System.out.println("\nConverted Values:");
        System.out.println("Bytes: " + converter.toBytes());
        System.out.println("Kilobytes: " + converter.toKB());
        System.out.println("Megabytes: " + converter.toMB());
        System.out.println("Gigabytes: " + converter.toGB());
        System.out.println("Terabytes: " + converter.toTB());
    }
}
```

## Output

***Sample Run 1***

```
Enter value: 2048
Enter unit (Bytes/KB/MB/GB/TB): KB
Converted Values:
Bytes: 2097152.0
Kilobytes: 2048.0
Megabytes: 2.0
Gigabytes: 0.001953125
Terabytes: 1.9073486328125E-6
```

***Sample Run 2***

```
Enter value: 5
Enter unit (Bytes/KB/MB/GB/TB): GB
Converted Values:
Bytes: 5.36870912E9
Kilobytes: 5242880.0
Megabytes: 5120.0
Gigabytes: 5.0
Terabytes: 0.0048828125
```

***Sample Run 3***

```
Enter value: 1500
Enter unit (Bytes/KB/MB/GB/TB): Bytes
Converted Values:
```

```
Bytes: 1500.0
Kilobytes: 1.46484375
Megabytes: 0.001430511474609375
Gigabytes: 1.3969838619232178E-6
Terabytes: 1.3642420526593924E-9
```

## Result

Thus the Java program using user-defined packages to convert data sizes was executed successfully and the output was verified.